JOSE DAVID RODRIGUEZ PINILLA

Full Stack Developer & IoT | Systems Engineering and Telematics Engineering Student — Universidad ICESI

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Available in Cali: on-site · hybrid · remote

PROFESSIONAL SUMMARY

Dual-degree student (Systems Engineering and Telematics Engineering) at Universidad ICESI, passionate about software development, systems architecture and the Internet of Things (IoT). Hands-on experience building versatile technology solutions: from web interfaces and applications with Java (Spring Boot) and frontend technologies (HTML, CSS, JavaScript, React), to end-to-end IoT systems integrating data-acquisition hardware (ESP32), real-time telemetry over MQTT, and robust backends with Python (FastAPI), SQL databases (PostgreSQL) and Docker containers. In telecommunications I work with Cisco Packet Tracer and MATLAB. Active member of IEEE ICESI, ComSoc and the Astrobiology Research Group. Looking for full-stack, backend or connected-systems opportunities where I can contribute analytical and technical value, with the ability to solve technical problems autonomously. English level: B1.

TECHNICAL SKILLS

Languages: Java · Python · SQL · JavaScript · C++ (ESP32/PlatformIO) · R

Backend & Databases: Spring Boot · FastAPI · SQLAlchemy · PostgreSQL · InfluxDB

Frontend & Web: HTML · CSS · JavaScript · React · Wix

Telecommunications & Networking: Cisco Packet Tracer (IPv4/IPv6 subnetting, routing protocols, VLANs) · MATLAB · MQTT

IoT, Data & Automation: ESP32 · Node-RED · Scikit-Learn · Automation Anywhere (bootcamp)

Tools: Docker · Docker Compose · Git

WORK EXPERIENCE

Front-End & Web Development Intern *Jun 2026 – Aug 2026*

Fundación Algo en Común — Part-time · Cali, Valle del Cauca · Hybrid

- Led the redesign, development and updating of the institutional web platform, optimizing the user interface and navigation experience to boost the organization's digital impact.
- Managed page configuration, responsive layout design and visual component optimization to ensure optimal performance across multiple devices.
- Collaborated on digital content administration and technical support for the websites backing the foundation's communication initiatives (Wix).

Sales Assistant *Dec 2022 – Feb 2023 and Dec 2023 – Feb 2024*

Calipieles — seasonal contract

- Customer service and cash register handling on seasonal contracts (2 seasons).

Laboratory Assistant *Aug 2022 – Jan 2023*

Fablab Cali, Universidad Autónoma de Occidente

- Assisted students and faculty; managed and maintained equipment in the digital fabrication lab.

Administrative Assistant *Mar – May 2022*

Center for Academic Excellence, Universidad Autónoma de Occidente

- Administrative support and assistance to students and faculty.

FEATURED ACADEMIC PROJECTS

Capstone Project 1 — IoT Tremor Measurement System (Archimedes Project) – Github URL *Feb – Jun 2026*

Individual project — Universidad ICESI · Repository: ProyectoIntegrador1-Tel

- Designed and implemented a complete IoT ecosystem to capture motion and pressure data during the clinical Archimedes spiral test, integrating MPU6050, LSM303DLHC and FSR402 sensors on an ESP32 (I2C / 12-bit ADC) sampling at 100 Hz.
- Built an asynchronous telemetry pipeline over MQTT with JSON batch transmission to prevent data loss on the local network.
- Developed the backend with FastAPI + PostgreSQL (SQLAlchemy ORM), featuring JWT/bcrypt authentication and role-based access control (Admin, Doctor, Patient).

- Designed a React frontend with real-time visualization of oscillations and patient grip force.
- Containerized the system's 3 layers (Nginx, API, database) with Docker Compose for reproducible, single-command deployment.

Systems and Communications I — IoT Monitoring and Agro-Climate Analytics – Github URL *Feb – Jun 2026*

Individual project — Universidad ICESI

- Developed an end-to-end IoT monitoring system for predictive analysis and anomaly detection in coffee and oil palm crops.
- Simulated climate telemetry (temperature, humidity, precipitation, pH) over MQTT and built a Node-RED processing gateway with agronomic range validation, persisting data in InfluxDB.
- Implemented a hybrid predictive model combining incremental learning (PassiveAggressiveRegressor) with periodic retraining (HistGradientBoostingRegressor), weighting predictions by historical MAE.
- Built a real-time visualizer (Matplotlib) with threshold-based anomaly detection.

EDUCATION

B.Sc. Systems Engineering — Universidad ICESI *Jan 2024 – Oct 2029 (exp.)*

Currently in 6th semester. Focus on software development, backend architectures, full-stack development and IoT solutions, with hands-on practice in Java, Spring Boot, Python, SQL and modern frontend technologies.

B.Sc. Telematics Engineering — Universidad ICESI *Jan 2024 – Feb 2029 (exp.)*

Currently in 6th semester. Focus on networking architecture, IoT systems and telecommunications: network design and configuration with Cisco Packet Tracer (IPv4/IPv6, routing, VLANs), MQTT telemetry, Docker containers and MATLAB for modeling and analysis.

Automation Bootcamp — Comfandi Automation Seedbed & Universidad ICESI *July 2026*

Automation Anywhere, cybersecurity, Lean principles and information security.

Programming Fundamentals — Universidad Tecnológica de Pereira *2021*

Technical High School Diploma in Commerce — I.E. Técnica de Comercio Simón Rodríguez *2021*

SENA Certifications in research and laboratory work *2018 – 2020*

ACTIVITIES & INVOLVEMENT

- IEEE member, ICESI Student Branch, since 2024. Represented the branch at the 2026 IEEE National Student Branch Meeting (RNR) in Rionegro, where the branch received multiple awards.
- Active participant in the ICESI Astrobiology Research Group since 2025.
- Member of the ComSoc ICESI student chapter since 2025.

SOFT SKILLS

Teamwork, technical leadership, interdisciplinary collaboration, proactivity, independent technical problem-solving, attention to detail and results-oriented mindset.

PERSONAL REFERENCES

- Robeiro Giraldo — Manager, Calipieles — 3206126523
- Abel Guzmán — Coordinator, IETC Simón Rodríguez — 3113672539